



ELIT Client Guide

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Introduction

Terminology and Acronyms

Term	Definition
ASCII	American Standard Code for Information Interchange – a standard developed by the American National Standards Institute (ANSI) for the numeric representation of alphanumeric and special characters.
AUTH SSL	An FTP command that negotiates an explicit secure connection.
COMPORD	EDI standard for the steel industry.
Dataset Name	The equivalent to a file name on an IBM mainframe computer.
DNS	Domain Name Server
DSN	Data Set Name, the equivalent to a file name on an IBM mainframe computer.
EBCDIC	Extended Binary Coded Decimal Interchange Code – numeric code developed by IBM for the representation of alphanumeric and special characters.
EDI	Electronic Data Interchange – electronic application-to-application exchange of structured business data between companies.
EDI Data	Text files formatted according to an industry standard such as EDIFACT, X12 ISA, ICS, BG, COMPORD, VICS, or TRADACOM.
EDIFACT	Electronic Data Interchange for administration, commerce, and transport. ISO standard for EDI.
ELIT MFT ID	A unique code that identifies a user in the ELIT MFT network, i.e., an ELIT MFT User ID. User ID, Mailbox, and Location Code are synonymous.
ELIT MFT	DXC Technology's Managed File Transfer
ELIT MFT Filename	A name from 1 to 10 characters assigned by the Sender that identifies the transfer operation within ELIT MFT. This name is often dictated by the recipient to identify the receiving application or the specific transmission.
FTP	File Transfer Protocol
GUI	Graphical User Interface
IP	Internet Protocol
Location Code	A unique code that identifies a user in the ELIT MFT network, i.e., an ELIT MFT User ID. User ID, Mailbox, and Location Code are synonymous.
Mailbox	Identifies a holding place for files sent to your ELIT MFT ID. A mailbox ID is equivalent to an ELIT MFT ID or Location Code.
Mainframe	IBM Multiple Virtual Storage Mainframe
Obtainable Files	Files in your mailbox waiting to be retrieved.

PCI DSS	PCI DSS stands for Payment Card Industry (PCI) Data Security Standard (DSS). It was developed by the major credit card companies as a guideline to help organizations that process card payments prevent credit card fraud, hacking and various other security issues. A company processing, storing, or transmitting credit card numbers must be PCI DSS compliant or they risk losing the ability to process credit card payments. Merchants and Service Providers must validate compliance with an audit by a PCI DSS Qualified Security Assessor (QSA) Company..
ReceiverID	ELIT MFT ID of the receiver of a file.
RFC	Request for Comments. A document in which an Internet standard or protocol is published.
SenderID	ELIT MFT ID of the sender of a file.
SSL	Secure Sockets Layer
Switcher	An ELIT MFT ID mailbox that can send and receive files on behalf of one or more other ELIT MFT IDs.
TCP/IP	Transmission Control Protocol/Internet Protocol
*THS/THD/TTR	ELIT MFT proprietary envelope records that contain file routing information.
Trading Partner	The sending and/or receiving parties involved in exchanging electronic data.
VAN	Value-Added Network
VPN	Virtual Private Network
X12	ANSI standard for EDI.
z/OS	z/OS (or zero-downtime/OS) is a 64-bit server operating system from IBM. It is the successor to the IBM mainframe operating system OS/390, combining MVS and Unix System Services (a POSIX-compliant mainframe implementation of UNIX formerly known as MVS Open Edition, or OpenMVS).

Trademarks

All trademarks or servicemarks used herein are the properties of their respective owners.

Overview

The ELIT platform independent client consists of two separate components written in the Java programming language: ELIT Send and ELIT Receive. The software will run on any platform that supports the Java version 8 or greater environment. The product supports Secure Socket Layer file transmissions with the world wide ELIT network. This provides a managed, secure, and auditable transport for the transfer of data between computer systems.

The current version of ELITClient communicates securely with the ELIT hub using direct or indirect (https/ftp) methods. By default, the ELITClient is configured for indirect communication.

Requirements for Indirect Communication:

Utilizes MFT protocol servers to provide encryption and additional routing.

For indirect, connectivity is required to MFT FTP server at ports 21 or 990 and ftp passive data ports 20900-20999.

Requirements for Direct Communication:

Allows ELITClient to send/receive files directly from MFT Hub.

Direct method is configured by specifying the keyword “direct” (direct=true/false) in ElitClient.properties file, which replaces old keyword “ssl”.

Direct method can be used only with MFT midrange servers – mft.svcs.entsvcs.com or mft2.svcs.entsvcs.com.

For direct communication connectivity to MFT server at port 25101 is required.

This document assumes the user has a basic understanding of Network protocols and the platform on which the Client will be run. (e.g. Windows, DOS, Unix).

Obtaining and Installing the ELIT Client Software

Step	Description
1	Download the ELIT Client software from the DXC public GitHub repository: https://github.dxc.com/pages/Global-BES/BES/mft_elit/ or contact the MFT Services team mft.services@dxc.com There is a downloadable file for the Microsoft Windows platform (ElitClient.zip) and there is one for the Unix platform (ElitClient.tar).
2	To install the ELIT Client, download the applicable file you're your platform and unzip or un-tar the files to the folder of your choice.
3	Once you have unzipped or un-tar'd the files update the scripts and properties files according to your needs. The scripts and properties files are described in this document.
4	The ELIT Client configuration files are the same for the Windows or Unix platforms except for the script related parms. The script files (e.g. bat, sh) are platform dependent.

Adding your user ID to ELIT

Contact the MFT services team (mft.services@dxc.com) and let them know that you wish to communicate with MFT using the ELIT Client.

They will:

- Define a User ID for your location.
- Define initial passwords for sending and receiving files.
- Need to know the TCP/IP address or DNS name of the machine where your ELIT Client will be running.

Need to know the TCP/IP port that your ELIT Client will listen on to receive files. By default the ELIT Client listens on port 25500. The provided scripts specify port 25400 to show how to override the default port. You can change this if you need to.

Contact Information

For additional information or Help Contact: mft.services@dxc.com.

Receiving Files

Overview

The ELITReceive component is used to receive file transmissions from other nodes in the ELIT network. This task should be started when the operating system is initialized. The software will listen on a predefined TCP/IP socket for connections from an ELIT server. Each new connection will run in a separate thread to allow for simultaneous file transfers. The software provides the option to rename the file and to initiate a user-defined process upon successful reception of the data. The ELITReceive program gets its configuration parameters from the "ElitClient.properties" file, located in the ELITReceive directory.

The ELITReceive component can also run as a Window's service on a Microsoft Window's machine. See [Appendix B](#).

ELIT Client Properties

This file will reside in the JVM's ClassPath and consists of the following options:

Option	Definition
userid	The userid that identifies this node to the ELIT network.
failmail	Optional E-Mail ID to which all file delivery errors are sent. Failmail feature by default utilizes ELIT MFT mail forwarders. To utilize this feature outside of the ELIT MFT network one must override failmail forwarder with – Dforwarder=my.mail.forwarder.com Java option. This option must be inserted into scripts starting ELIT receive and/or ELIT send programs: elit and elitsend.
sh	Optional global shell used to execute scripts when a file is successfully received.
directory	Optional global directory to place files into.
dsn	Optional new data set name for all files received.
lpaccept	Optional global /path/file to define IP addresses allowed to connect to this node. (IP addresses must be on separate lines in named file)
script	Optional script to execute for all files received.
LogScriptStdout	True/False to specify that stdout messages from the executed script should be written to the log4j log. Default is false.
logScriptStderr	True/False to specify that stderr messages from the executed script should be written to the log4j log. Default is false.
QuoteParms	Optional flag if set to true will enclose all script parameters in double quotes. Use this if script parameters contain special characters that cause errors in script processing. The script will then have to remove the quotes from each parameter.

Option	Definition
FileNameScrub	Optional parameter to replace unwanted characters or a sequence of characters expressed with regular expression with a given replacement in a filename. Needs to be define as regex=replacement. For example, [:\$!]=_ will replace any occurrences of colon, dollar sign or exclamation mark characters with an underscore character.
direct	Optional parameter to indicate that ELIT Client will be receiving data directly from ELIT MFT hub, bypassing MFT protocol server. Default value is false.
keystore	Optional parameter to specify non-default keystore to be used for TLS communication. The default keystore contains a self-signed x.509 certificate trusted in ELIT network and by ELIT Client. For installations requiring CA issued certificate this parameter can be used to override the default settings.
keystorepswd	Obfuscated password for keystore. This is only used if keystore parameter is active. Obfuscate utility must be used to obfuscate the password.
FileName.sh	Optional shell used to execute a script when any data using this "FileName" is successfully received.
FileName.dir	Optional directory to place files using this "FileName".
FileName.dsn	Optional new data set name for a file received using this "FileName".
FileName.script	Optional script to execute when data using this "FileName" is successfully received.
FileName.logScriptStdout	True/False to specify that stdout messages from the script executed by this "FileName" should be written to the log4j log. Default is value specified by the logScriptStdout option.
FileName.logScriptStderr	True/False to specify that stderr messages from the script executed by this "FileName" should be written to the log4j log. Default is value specified by the logScriptStderr option.

The FileName.sh, .dir and .dsn will override the global sh, directory and dsn entries when both are specified.

Both the "directory" and "FileName.dir" parameters allow the following comma separated processing directives after the directory name:

1. lc - Lower Case the original Data Set Name.
2. -x - Remove the upper most "x" nodes from the original Data Set Name.
3. + - Append the ELIT MFT "FileName" as the lowest level of the directory.
4. Sb - Forward the file to the SeeBeyond interface.

When executing a script after the successful reception of a file the following elements are passed as positional parameters:

1-Full Data Set Name including directory, 2-SenderId, 3-ReceiverId, 4-FileName, 5-Reference #, 6-Original Data Set Name, 7-File Control #

The “directory,” “dsn,” “FileName.dir,” and “FileName.dsn” can optionally contain any of the following symbolic parameters for further customization of the output Data Set Name.

Parameter	Definition
\$dat	Date the file was received in the format: yyyy_mm_dd.
\$tim	Time the file was received in the format: hh_mm_ss.
\$rid	ELIT location code of the receiver.
\$sid	ELIT location code where the file originated.
\$ref	ELIT file “Reference Number”.
\$fnm	ELIT “FileName”.
\$fcn	ELIT “File Control Number”.
\$dsn	Original Data Set Name.

Example ELITClient.properties file for the Windows platform

```
#
#   ELIT Java Client Properties File
#
userid          = RICKP
failmail        = James.Bell@dxs.com
sh              = CMD /C
direct          = false
directory       = d:/ElitClient/Received
dsn             = $dat.$tim.$sid.$dsn
ipaccept        = c:/ElitClient/ELITReceive/ipaccept.txt
logScriptStdout = false
logScriptStderr = true
#script         = c:/ElitClient/ELITReceive/elitfile.bat
```

Example ELITClient.properties file for the Unix platform

```
#    ELIT Java Client Properties File
#
userid          = RICKP
failmail        = James.Bell@dxs.com
sh              = /bin/bash
direct          = true
directory       = ./ElitClient/Received
dsn             = $dat.$tim.$sid.$dsn
ipaccept        = ./ipaccept
logScriptStdout = false
logScriptStderr = true
#script         = ./ElitClient/ELITReceive/elitfile.bat
#TESTFILE.script = ./ElitClient/ELITReceive/elitfile.bat
#TESTFILE.dsn    = $dat.$tim.literal.$sid.$rid.$ref.$fnm.$fcn.$dsn
#TEST2.script    = ./ElitClient/ELITReceive/missing.bat
```

Creating the encrypted receive password

The password associated with the userid defined in the ElitClient properties file is encrypted and stored in a file named passwd.<userid> as shown below. The password file resides in the same directory as the ElitClient properties file. A password reset utility is provided for setting the initial password and for resetting the password if necessary.

Windows password reset bat file

A Windows batch file for resetting the ElitReceive password is provided in the file passwdrecv.bat.

Unix password reset script

A Unix shell script for resetting the ElitReceive password is provided in the file passwdrecv.sh.

Setting the initial password for ElitReceive

Elit will login to ElitReceive using the userid and password assigned by MFT Services. The assigned userid and initial password must be defined before running the ElitReceive component for the first time. The userid is defined as a property in the ElitClient.properties file.

Once the userid property has been defined, run the password reset utility appropriate for your platform. The password reset utility will read the ElitClient.properties file to determine the userid and will then prompt for a password as shown below. This example is for userid abc:

```
D:\ElitClient\ELITReceive> passwdrecv.bat  
Enter password for userid abc: *****  
  
Storing new password in  
D:\ElitClient_JDK1.5\ELITReceive\passwd.abc  
Password reset successful, (re)start Elit Client to begin using  
new password
```

The new password is encrypted and written to the file passwd.abc. Once the userid and password have been defined, ElitReceive can be started and will be able to receive files from ELIT.

For non-direct configurations (direct = false) passwords assigned by MFT Services are "single-use" passwords that will be automatically changed by ELIT during the next login. Passwords will also be automatically changed by ELIT upon expiration. This password change process is transparent to you and enhances security by assuring that the password is not known by a third party.

TLS (Transport Layer Security, a.k.a SSL) keystore

ELIT Client includes a default keystore for secure communication with ELIT servers. The default keystore contains a self-signed x.509 certificate trusted by ELIT Client and ELIT Servers. This certificate will expire on January 1, 2040. If security requirements mandate that ELIT Client Receive component uses a CA (Certificate Authority) issued certificate, ELIT Client can be configured to override the use of the default keystore with one provided from an external environment. ElitClient.properties' keystore and keystorepswd parameters can be used to point to an external keystore and provide the password for it. Value for keystorepswd needs to be obfuscated with provided obfuscate utility. Java's keytool utility can be used to create a new keystore, generate CSR (Certificate Signing Request) and import the CA certificate.

Running the ElitReceive component

To start the ELITReceive component, execute the ELITReceive startup script unless you are running the ELITReceive Component as a Window's Service. See Appendix B.

ELITReceive logs messages to the file ElitReceive.log in the logs directory. By default, log files rollover each day and are kept for 14 days. ElitReceive by default uses Apache log4j logging services. Logging configuration parameters are defined in the file log4j2.xml and log4j2Console.xml. Do not edit these files unless you need to change the logging configuration for some reason and understand how log4j works. It is possible to implement a different logging services other than log4j services. slf4j-jdk14-1.7.32.jar is supplied with the ELITClient for java logging. slf4j-impl.jar symbolic link can be replaced with slf4j-jdk14-1.7.32.jar and appropriate java logging properties file has to be implemented for adopting java logging.

Example Windows bat file startup script elit.bat

```
@echo off
set cpath=..\SymLinks\ElitClient.jar
set cpath=%cpath%;..\SymLinks\FTPBean.jar
set cpath=%cpath%;..\SymLinks\ElitUtil.jar
set cpath=%cpath%;..\SymLinks\mail.jar
set cpath=%cpath%;..\SymLinks\activation.jar
set cpath=%cpath%;..\SymLinks\slf4j-api.jar
set cpath=%cpath%;..\SymLinks\slf4j-impl.jar
set cpath=%cpath%;..\SymLinks\log4j-api.jar
set cpath=%cpath%;..\SymLinks\log4j-core.jar
set cpath=%cpath%;.

java -classpath %cpath% ^
    -DELITPASV=high ^
    -DCryptoLibrary=../ObjRepos/keys ^
    -DMFTX509TrustManager.throwException=true ^
    ElitClient.ElitClientServ 25500 ^
1>logs\ElitReceiveStdOut.log 2>logs\ElitReceiveStdErr.log
```

Example of Unix file startup script elit.sh

```
#!/bin/sh
#
cpath=./SymLinks/ElitClient.jar
cpath=$cpath':../SymLinks/FTPBean.jar'
cpath=$cpath':../SymLinks/ElitUtil.jar'
cpath=$cpath':../SymLinks/mail.jar'
cpath=$cpath':../SymLinks/activation.jar'
cpath=$cpath':../SymLinks/slf4j-api.jar'
cpath=$cpath':../SymLinks/slf4j-impl.jar'
cpath=$cpath':../SymLinks/log4j-api.jar'
cpath=$cpath':../SymLinks/log4j-core.jar'
cpath=$cpath':..'

echo $$ > ElitReceiveShell.pid

java -DELITPASV=high -classpath $cpath \
      -DCryptoLibrary=../ObjRepos/keys \
      -DMFTX509TrustManager.throwException=true \
      ElitClient.ElitClientServ 25500 \
      1>logs/ElitReceiveStdOut.log2>logs/ElitReceiveStdErr.log&
echo $! > ElitReceiveJava.pid
```

Running the ElitReceive component

To shut down the ELITReceive Component execute shut.bat (Windows) or shut.sh (Unix).

Example Windows shut.bat

```
@echo off

set cpath=..\SymLinks\ElitClient.jar
set cpath=%cpath%;..\SymLinks\ElitUtil.jar
set cpath=%cpath%;..\SymLinks\mail.jar
set cpath=%cpath%;..\SymLinks\activation.jar
set cpath=%cpath%;..\SymLinks\slf4j-api.jar
set cpath=%cpath%;..\SymLinks\slf4j-impl.jar
set cpath=%cpath%;..\SymLinks\log4j-api.jar
set cpath=%cpath%;..\SymLinks\log4j-core.jar
set cpath=%cpath%;.

java -classpath %cpath% ^
    -DMFTX509TrustManager.throwException=true ^
    -DCryptoLibrary=../ObjRepos/keys ^
    ElitClient.ElitClientShut 25500
```

Example Unix shut.sh

```
#!/bin/sh
#
cpath=../SymLinks/ElitClient.jar
cpath=$cpath'../SymLinks/ElitUtil.jar'
cpath=$cpath'../SymLinks/mail.jar'
cpath=$cpath'../SymLinks/activation.jar'
cpath=$cpath'../SymLinks/slf4j-api.jar'
cpath=$cpath'../SymLinks/slf4j-impl.jar'
cpath=$cpath'../SymLinks/log4j-api.jar'
cpath=$cpath'../SymLinks/log4j-core.jar'
cpath=$cpath'..'

java -classpath $cpath \
    -DMFTX509TrustManager.throwException=true \
    -DCryptoLibrary=../ObjRepos/keys \
    ElitClient.ElitClientShut 25500
```

Sending Files

Overview

The ELITSend component is used to transmit files to other nodes in the ELIT network. A separate task is initiated for each transmission. Multiple files can be serially sent in a single task, and multiple instances of the software can run simultaneously. The basic syntax of the command structure is very similar to the standard FTP specification. The product does include added retry and notification commands for a more robust implementation.

ELITSend Component commands

ELIT Send Component Commands are specified in a parameter file that is passed to the ELITSend Component as a parm of the ELITSend startup script.

The following commands can be included in the ELITSend parameter file (elit.parms).

Parameter	Definition
options	Specifies the jobname, retry levels, mail, and transmission type.
site	Specifies the site parameters.
elit/ftp	Specifies the hostname and port of the receiving ELIT server. The "elit" parameter is used for a secure (SSL) direct transmission to ELIT hub using the proprietary ELIT protocol. The "ftp" parameter along with a port prefix of 'N' or 'S', is used to send the data stream using SSL encryption to an ELIT Secure FTP Server. The port prefix 'N' specifies that the security protocol will be explicitly negotiated with ELIT. This is commonly known as Explicit FTPS (FTP/SSL) or AUTH TLS and is the preferred secure option. (e.g. mft.svcs.entsvcs.com:N21) The port prefix 'S' specifies implicit session security. This is commonly known as Implicit FTPS (FTP/SSL). (e.g. mft.svcs.entsvcs.com:S990)
user	Userid identifying this node to the ELIT network.
cd	Change Directory command used to specify routing information.
put	Identifies the file to transmit and optionally the ELIT routing information.

pwwpath	Identifies the path to the ElitSend password file for the specified user.
---------	---

The “options” statement allows several blank delimited keywords to customize the transmission processing.

jobname=xxxx	Specifies the name to be used to identify this transmission in the optional E-mails.
failmail=xxxx	Specifies the E-mail ID(s) to receive notification of a total failure of this jobstream. Failmail feature by default utilized DXC’s mail forwarders. To utilize this feature outside of ELIT MFT network one must override failmail forwarder with – Dforwarder=my.mail.forwarder.com Java option. This option must be inserted into scripts starting ELIT receive and/or ELIT send programs: elit and elitsend
binary	Indicates that the file(s) should be sent in binary format (the default is text format).
retryw=x:y:zzz	The “w” is the level of the retry. The program will accept up to three levels of retry information. The “x” is the number of seconds between retry intervals. The “y” is the number of intervals at this level. The optional “zzz” is the E-mail ID(s) to receive notification of a failure at this level.

Note: Most alpha paging services have an E-mail component that allows this software to page the system administrator when a file transmission fails.

In order to initiate a delivery request, the user must identify the recipient of the data. This can be accomplished using following different methods.

The CD Command

The “CD” command can be used to specify the SenderId, ReceiverId, ELIT Filename, Reference #, and data format. The SenderId defaults to the user specified on the “USER” command. If a unique SenderId is required, it is supplied as the first token, and must be prefixed with a slash. Missing parameters should be specified with a period (“.”) as a placeholder. The command syntax follows:

```
cd /SenderId/ReceiverId/ELIT-Filename/RefNo/KEEPLF
```

Examples of the CD command:

```
cd /user2/user1/aaadata
```

The next put command will deliver the data specified to “user1” from

"user2", with an ELIT filename of aaadata.

```
cd
user1/xyzbilling/day33
```

The next put command will deliver the data specified to "user1" from the location specified on the "USER" command, with a ELIT filename of "xyzbilling", and a reference number of "day33".

```
cd
alk/abcinvoice/./keeplf
```

The next put command will deliver the data specified to location "a1k", with a ELIT filename of "abcinvoice", and a reference number of "." (Placeholder). The "keeplf" flag only has meaning when sending a file to a MVS system in ascii mode. This flag indicates that the data should be translated from ascii to ebcdic, but should not be stripped of carriage control or line feeds. This option is usually used when transmitting EDI documents from an ascii based system to MVS.

The SITE Command

Another method of specifying the routing information is with the "SITE" command. The "SITE" command will accept comma or blank delimited keyword/value pairs for the required information. The available keywords are:

```
SID=SenderId,RID=ReceiverId,FNM=ELIT-
Filename,REF=RefNo,STRIPLF/KEEPLF
```

Examples of the SITE command:

```
site
sid=user2,rid=user1,fnm=aaadata
```

The next put command will deliver the data specified to "user1" from "user2", with a ELIT filename of aaadata.

```
site
rid=user1,fnm=xyzbilling,ref=day33
```

The next put command will deliver the data specified to "user1" from the location specified on the "USER" command, with a ELIT

filename of
"xyzbilling", and a
reference number of
"day33".

```
site sid=xyz,rid=fromdata,keeplf
```

The next put command will deliver the data specified to the recipient id specified in the EDI data header from the sender id specified in the EDI data header, with a filename specified in the EDI data header, if designated. Keeplf indicates that this data is EDI and not TEXT.

The PUT Command

The routing information can also be included in the "PUT" command by specifying the full directory structure on the output file. The syntax would follow the same rules as the "CD" command. The ELIT software can also extract the routing information from EDI data files. The reserved ReceiverId of "FROMDATA" will initiate this process. The routing information found in the EDI document controls the transmission.

Example of the Windows Elitsend parameter file (elit.parms)

```
options jobname=RAPsJob
options retry1=5:5:youremail@yourcompany.com
options retry2=10:10:youremail@yourcompany.com
options failmail=youremail@yourcompany.com

ftp mft2.svcs.entsvcs.com:N21
user RICKP
cd LAM/rapsfile/99999
pwwpath \ElitClient\ElitSend\
put d:\ElitClient\Send\helloWorld.txt
```

Example of the Unix Elitsend parameter file (elit.parms)

```
options jobname=senddata binary
options retry1=5:5: youremail@yourcompany.com
options retry2=60:5: youremail@yourcompany.com
options retry3=600:5: youremail@yourcompany.com
options failmail=youremail@yourcompany.com
elit mft.svcs.entsvcs.com
user PAULS
cd $2./$3.
put $1.
```

Example of the Windows startup script for elitsend (elitsend.bat)

```
@echo off

set cpath=..\SymLinks\ElitClient.jar
set cpath=%cpath%;..\SymLinks\FTPBean.jar
set cpath=%cpath%;..\SymLinks\ElitUtil
set cpath=%cpath%;..\SymLinks\mail.jar
set cpath=%cpath%;..\SymLinks\activation.jar
set cpath=%cpath%;..\SymLinks\slf4j-api.jar
set cpath=%cpath%;..\SymLinks\slf4j-impl.jar
set cpath=%cpath%;..\SymLinks\log4j-api.jar
set cpath=%cpath%;..\SymLinks\log4j-core.jar
set cpath=%cpath%;.

java -classpath %cpath% ^
    -DELITPASV=high ^
    -DCryptoLibrary=../ObjRepos/keys ^
    -DMFTX509TrustManager.throwException=true ^
    ELITSend.ELITSend elit.parms %1 %2 %3 ^
    1>logs\ElitSendStdOut.log 2>logs\ElitSendStdErr.log
```

Example of the Unix startup script for elitsend (elitsend.sh)

```
#!/bin/sh
#
#      Send a File to a remote location via ELIT
#
cpath=./ElitClient.jar
cpath=$cpath':./FTPBean.jar'
cpath=$cpath':./ElitUtil.jar'
cpath=$cpath':./mail.jar'
cpath=$cpath':./activation.jar'
cpath=$cpath':./slf4j-api.jar'
cpath=$cpath':./slf4j-impl.jar'
cpath=$cpath':./log4j-api.jar'
cpath=$cpath':./log4j-core.jar'
cpath=$cpath':.'

java -classpath $cpath \
    -DELITPASV=high \
    -DMFTX509TrustManager.throwException=true \
        ELITSend.ELITSend elit.parms $1 $2 $3
1>SendClient.log 2>SendDebug.log
```

Creating the encrypted send password

The password required to login to ELIT using the ELITSend process is encrypted and stored in a specific password file. The initial password provided by MFT Services must be used in the “passwdsend.bat” or “passwdsend.sh” utility provided.

A user’s password may be manually reset at any time, but must be coordinated with the associated ELIT User id to prevent “out-of sync” conditions. Contact the ELIT Service Desk to coordinate a password change situation.

See Appendix H for an explanation of PCI DSS V1.1 password policies.

Running passwdsend.bat (Windows) or passwdsend.sh (UNIX) utility

The passwdsend utility accepts 3 positional parameters, UserID, Password, and Password Path. The UserID is used to associate the generated password with a specific ELIT user id and to create or update a unique password file (passwd.<UserID>) for this particular user. It must be the same as your assigned ELIT user id and can be 3 -17 characters in length.

The password provided (as initially specified by the MFT Services group, or coordinated with the MFT Service Desk), will be encrypted and stored in the user’s unique password file. It must be 8 alphanumeric characters in length.

The Password Path parameter is used to store the user's unique password file in the desired directory. It should only specify the PATH of where the generated user's password file should be created.

The UserID, and Password Path parameters must also be passed in the elit.parm file during ELITSend execution so that the encrypted password will be read correctly during session establishment with ELIT.

See Appendix H for an explanation of PCI DSS V1.1 password policies.

Automatic Password Updates

The ELITSend process now automatically stores, encrypts, decrypts and generates new passwords when required, based on messages received while communicating indirectly with ELIT, via FTP Server. During secure communications with ELIT, if a user's password has expired or is about to expire the ELITSend process will generate and encrypt a new password to be used on the next subsequent secure session with ELIT. Once ELIT accepts this new password, ELITSend will mark it as the current password and use it until such notification occurs again. Should ELIT reject the new password, ELITSend will automatically generate a different password and retry on the next secure session with ELIT. This process should eliminate the need for a manual password reset sequence to be completed whenever the current password expires in ELIT. As mentioned previously, a manual password reset sequence is available should a password "out-of-sync" condition occur or the user wishes to change the password before it expires. A manual password reset sequence must still be used when communicating with ELIT hub directly (direct = true). See above sections for specifics on manually resetting a password.

See Appendix H for an explanation of PCI DSS V1.1 password policies.

passwdsend.bat example for the Windows platform

```
@echo off
set cpath=..\SymLinks\ElitClient.jar
set cpath=%cpath%;..\SymLinks\FTPBean.jar
set cpath=%cpath%;..\SymLinks\ElitUtil.jar
set cpath=%cpath%;..\SymLinks\mail.jar
set cpath=%cpath%;..\SymLinks\activation.jar
set cpath=%cpath%;..\SymLinks\slf4j-api.jar
set cpath=%cpath%;..\SymLinks\slf4j-impl.jar
set cpath=%cpath%;..\SymLinks\log4j-api.jar
set cpath=%cpath%;..\SymLinks\log4j-core.jar
set cpath=%cpath%;.

java -classpath %cpath% ^
    -Dlog4j.configurationFile=log4j2Console.xml ^
    -DCryptoLibrary=../ObjRepos/keys ^
    ELITSend.ElitPassIn %1 %2 %3
```

passwdsend.sh example for the Unix platform

```
#!/bin/sh
cpath=../SymLinks/ElitClient.jar
cpath=$cpath'../SymLinks/FTPBean.jar'
cpath=$cpath'../SymLinks/ElitUtil.jar'
cpath=$cpath'../SymLinks/mail.jar'
cpath=$cpath'../SymLinks/activation.jar'
cpath=$cpath'../SymLinks/slf4j-api.jar'
cpath=$cpath'../SymLinks/slf4j-impl.jar'
cpath=$cpath'../SymLinks/log4j-api.jar'
cpath=$cpath'../SymLinks/log4j-core.jar'
cpath=$cpath'..'

java -classpath $cpath \
    -Dlog4j.configurationFile=log4j2Console.xml \
    -DCryptoLibrary=../ObjRepos/keys \
    ELITSend.ElitPassIn $1 $2 $3
```

Sending file with a directory monitor

Overview

Another option for sending files is the directory monitor capability of the ELITReceive component. A directory monitor will automatically send files that are placed in a defined directory. To enable directory monitoring define one to five monitored directories in the ElitClient.properties file. The ELITReceive component then monitors each defined directory continuously for files to send to the specified destination in addition to performing its normal receive function.

Defining a monitored directory

Monitored directories are defined in the ElitClient.properties file as follows:

Property	Definition
MonDir.<name>	Defines a monitored directory with the specified <name> and directory path followed by a list of comma-separated attributes. The following attributes are available with default values as specified: i=300 Poll interval in seconds r=<receiver> ELIT receiver ID a=\$DEFAULT ELIT application ID (optional) u=mft.svcs.entsvcs.com:N21 ELIT FTP server address:port o=binary FTP binary option (default is ascii) o=keep1f ELIT keep1f option (default is striplf) pwpath=<dir> Password file directory path adpath=<dir> Archive directory path (optional)

Example:

```
MonDir.lexicom=D:/ElitClient/MonDirs/lexicom,  
i=60,r=lexicom,a=INVOICE,u=mft.svcs.entsvcs.com:N21,o=binary,  
pwpath=D:/ElitClient/ElitSend,adpath=D:/ElitClient/MonDirs/arch  
ive
```

Directory paths are specified using the '/' character and are converted to the platform-specific character at runtime ('\ on Windows and '/' on Unix).

Defined directory monitors start when the ELITReceive component is started and run continuously. Status and error messages pertaining to directory monitors are written to the ELITReceive log. A directory monitor can be temporarily suspended if required by

renaming the directory (e.g., D:/ElitClient/MonDirs/lexicom.suspend). Monitoring will be resumed when the directory is renamed back to its original name.

When a directory monitor detects files in the directory it will connect to the specified ELIT FTP server and will login with the userid specified in ElitClient.properties. A maximum number of monitored directories is 25.

ELIT uses different passwords when sending and receiving files for a location code. This is why the ELITReceive and ELITSend components have different password files. These password files have the same name (passwd.<userid>) and care must be taken to set pwpath to the password file used by ELITSend if it exists. If you are not using ELITSend and have not defined a send password file for the userid, follow the steps defined in the ELITSend chapter for creating the initial password file and then set pwpath accordingly.

The directory monitor sends all detected files in the same FTP session and by default will delete each file after it has been successfully sent to ELIT. If you want to archive sent files, set adpath to the archive directory path. The directory monitor will then move files to the archive directory instead of deleting them. Archived files will be renamed to the original filename followed by the receiver ID, the current date and time stamp, and a ".sent" suffix (e.g., "file.lexicom.20071228152846707.sent").

Appendix A – ELIT filename conventions

The ELIT software identifies individual files by the following six data elements.

Data Element	Definition	Number of Characters
ELIT Filename	Specified by the user during data delivery. The ELIT filename can be used to trigger post processing at the destination host.	1–10
SenderId	The user or location of where the data originated.	3–17
ReceiverId	The user or location of where the data is to be delivered.	3–17
Reference #	Specified by the user during data delivery to further identify the data.	1–35
Control #	Assigned by the ELIT software to identify an individual file.	
DataSetName	The data set name of the file at the originating host.	1–44

These data elements are used to route files through the network and optionally initiate the processing of the data at the final destination.

Appendix B – Running ELITReceive as a Windows service

A **Windows service** is an application that starts when the Microsoft Windows operating system is booted and runs in the background as long as Windows is running. It is similar in concept to a Unix daemon. The ElitReceive component can be installed to run as a Windows service if desired. There are several utilities available on the Internet for installing applications like ElitReceive as a Windows service. JavaService is one example which is available at:

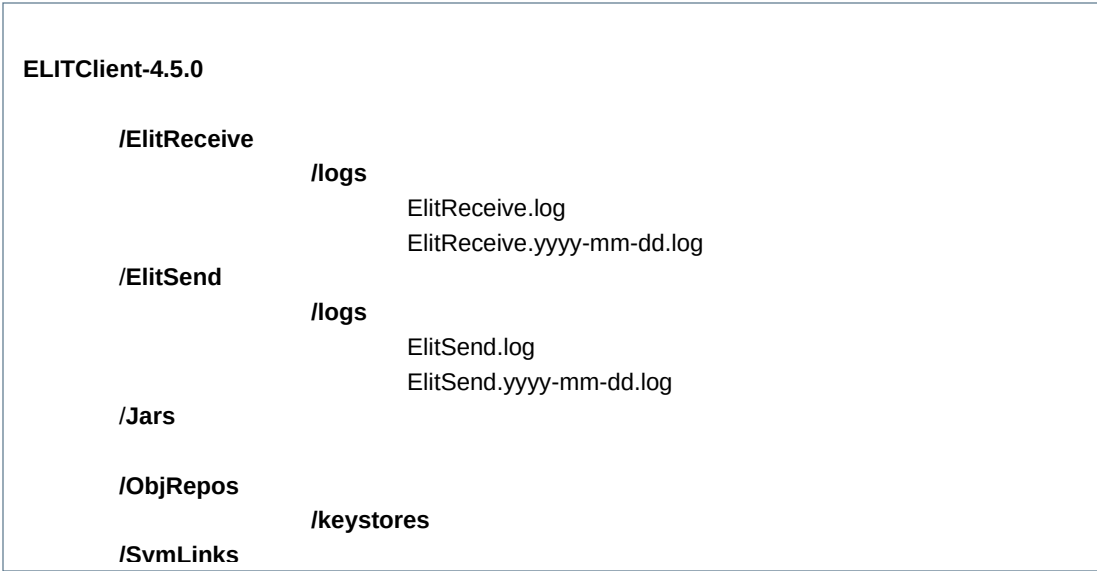
Home - Java Service Wrapper (tanukisoftware.com)

Appendix C – ELIT Client Directory Structure

Example of the ELITClient directory structure used with the Windows examples in this document.



Example of the ELITClient directory structure used with the UNIX examples in this document.



Appendix D – ELIT Log Files

The ELIT Client produces the following log files:

Daily log file

The daily log contains accumulated session messages for one day. It is produced by the application using log4j and written to the logs directory where the Java application is started. Log4j logging is controlled by the log4j2.xml and log4j2Console.xml files that should not be modified without an in-depth knowledge of log4j. Log4j logs are kept for 14 days, and then discarded.

Example:

ElitSend.yyyy-mm-dd.log

Standard output log file

The standard output log contains all messages written to standard output since the application was started and may be empty. This is the Java System.out.println log and is redirected to a file by the start up script with “1>” followed by the log name.

Windows Example:

```
java -classpath %CP% ELITSend.ELITSend elit.parms  
1>logs\ElitSendStdOut.log 2>logs\ElitSendStdErr.log
```

Unix Example:

```
java -classpath $classpath ELITSend.ELITSend elit.parms  
1>logs/ElitSendStdOut.log 2>logs/ElitSendStdErr.log
```

Standard error log file

The standard error log contains all messages written to standard error since the application was started and may be empty. This is the Java System.err.println log and is redirected to a file by the start up script with “2>” followed by the log name.

Windows Example:

```
java -classpath %CP% ELITSend.ELITSend elit.parms  
1>logs\ElitSendStdOut.log 2>logs\ElitSendStdErr.log
```

Unix Example:

```
java -classpath $classpath ELITSend.ELITSend elit.parms  
1>logs/ElitSendStdOut.log 2>logs/ElitSendStdErr.log
```

Appendix E – ELITSend Dynamic Parm File Modification

The parm file that ELITSend uses contains configuration options and commands that are read and executed by the ELITSend component. See ELIT Send Component Commands.

The ELITSend Component has a built in feature that allows you to dynamically modify the parm file used by the ELITSend component. This is accomplished by passing parameters on the ELITSend start up script file. These parameters are positional and replace associated variables in the parm file.

Using this technique you don't have to have many different parm files or have to constantly change the parm file for different files you need to transmit. You can change the contents of the parm file dynamically through the passing of positional parameters on the ELITSend start up script.

This is best understood by showing how it works in the following Example:

Template.parms (parameter file read by the ELITClient component)

```
options jobname=RAPsJob
options retry1=5:5:youremail@yourcompany.com
options retry2=10:10:youremail@yourcompany.com
options failmail=youremail@yourcompany.com

elit mft.svcs.entsvcs.com:25101
user ftpxxx/ftpxxx
cd $1./$2./99999

put d:\ElitClient\Send\helloWorld.txt
```

The above parameter file shows two variables that will be replaced by parameters passed into the SendClient.bat startup script.

The \$1. variable will be replaced by the first positional parameter on the SendClient.bat start up script. This is the SenderID in the parm file.

The \$2. variable will be replaced by the second positional parameter on the SendClient.bat start up script. This specifies the ELIT filename.

SendClient.bat start up script for the ELITSend component (Windows example)

```
@echo off
set cpath=..\SymLinks\ElitClient.jar
set cpath=%cpath%;..\SymLinks\FTPBean.jar
set cpath=%cpath%;..\SymLinks\ElitUtil
set cpath=%cpath%;..\SymLinks\mail.jar
set cpath=%cpath%;..\SymLinks\activation.jar
set cpath=%cpath%;..\SymLinks\slf4j-api.jar
set cpath=%cpath%;..\SymLinks\slf4j-impl.jar
set cpath=%cpath%;..\SymLinks\log4j-api.jar
set cpath=%cpath%;..\SymLinks\log4j-core.jar
set cpath=%cpath%;.

java -classpath %cpath% ^
    -DELITPASV=high ^
    -DCryptoLibrary=../ObjRepos/keys ^
    -DMFTX509TrustManager.throwException=true ^
    ELITSend.ELITSend elit.parms %1 %2 ^
    1>logs\ElitSendStdOut.log 2>logs\ElitSendStdErr.log
```

%1 and **%2** are replaced with the first and second positional parameters specified on the SendClient.bat file command.

SendClient.sh start up script for the ELITSend component (UNIX example)

```
#!/bin/sh
#
#    Send a File to a remote location via ELIT
#
cpath=./ElitClient.jar
cpath=$cpath':./FTPBean.jar'
cpath=$cpath':./ElitUtil.jar'
cpath=$cpath':./mail.jar'
cpath=$cpath':./activation.jar'
cpath=$cpath':./slf4j-api.jar'
cpath=$cpath':./slf4j-impl.jar'
cpath=$cpath':./log4j-api.jar'
cpath=$cpath':./log4j-core.jar'
cpath=$cpath':.'

java -classpath $cpath \
    -DELITPASV=high \
    -DMFTX509TrustManager.throwException=true \
    ELITSend.ELITSend elit.parms $1 $2
1>SendClient.log 2>SendDebug.log
```

\$1 and **\$2** are replaced with the first and second positional parameters specified on the SendClient.sh file command.

Note: If positional parameters start with “special characters”, place parameters in quotes.

Windows Example of executing the SendClient.bat file to start up the ELITSend component

```
C:\SendClient.bat LAM rapsfile
```

Unix Example of executing the SendClient.bat file to start up the ELITSend component

```
SendClient.sh LAM rapsfile
```

LAM is the first positional parameter and will replace %1(windows) or \$1(UNIX) in the bat or .sh file at execution time.

Rapsfile is the second positional parameter and will replace %2(windows) or \$2(UNIX) in the bat or .sh file at execution time.

The modified elit.parm file will look like this at execution time after its variables are substituted with positional parameters.

```
options jobname=RAPsJob
options retry1=5:5: youremail@yourcompany.com
options retry2=10:10: youremail@yourcompany.com
options failmail=youremail@yourcompany.com
elit mft.svcs.entsvcs.com:25101
user ftpxxx
cd LAM/rapsfile/99999put
d:\ElitClient\Send\helloWorld.txt
```

Appendix F – Site specific installation variables

Variable	Value
User ID for Receiving files from ELIT	<i>The User ID that identifies this node to the ELIT network when ELIT sends files to this node.</i>
Password for User ID used for receiving	<i>The password that is used with the Receiving User ID when ELIT sends files to this node.</i>
User ID for Sending files to ELIT	<i>The User ID that identifies this node to the ELIT network when this node sends files to ELIT.</i>
Password for User ID used for sending	<i>The password that is used with the Sending User ID when this node sends files to ELIT.</i>
Email address for receiving error messages	<u><i>youremail@yourcompany.com</i></u>
Host name of ELIT	<i>DNS of the ELIT region that this node communicates with.</i>
Port on ELIT	<i>Port number associated with the ELIT region DNS that this node communicates with.</i>

Appendix G – Startup script options

Running the Client from a non-privileged account

The default local port for passive data connections is 20. If you need to run the ELIT Client from a non-privileged account that is not permitted to bind sockets to well known ports (e.g. < 1024) you can use the java parm on the startup script –DELITPASV. - DELITPASV=high is a property that forces the ELIT Client to use a local port > 1023 when establishing passive data connections. A specific port can be also specified (e.g. ELITPASV=20123).

Passive IP address

When using FTP protocol Elit Client will by default establish passive FTP data connections using the same IP address that it used to connect to the Elit FTP server. Certain network configurations (e.g., involving load balancers) may instead require the Elit Client to use the IP address returned by the FTP server in the PASV command response. To use the PASV response IP address specify the java parameter **-Delit.ftpbean.use.pasv.ip=true** in the startup script.

Appendix H – PCI DSS V1.1 Password Information

The passwords for PCI DSS v1.1 must adhere to the following policies:

The system owner identity will be verified before performing password resets of the Location IDs. The location IDs will be assigned by a consistent identifier assignment. The identifiers will contain information such as name and related access information. (PCI DSS v1.1 8.5.2, EDSSCS0456, EDSSCS0467)

First-time passwords will be set to a unique value for each location ID and changed immediately after the first use. (PCI DSS v1.1 8.5.3)

Access for any terminated users associated to Location IDs will be immediately revoked. (PCI DSS v1.1 8.5.4)

Inactive Location IDs will be removed at least every 90 days. After 90 days of being disabled, the Location IDs will be removed from the Location ID table. (PCI DSS v1.1 8.5.5, EDSSCS0463)

There will be no group, shared, or generic accounts and passwords. (PCI DSS v1.1 8.5.8)

Location ID passwords will be changed at least every 90 days. (PCI DSS v1.1 8.5.9)

Location ID passwords will be 8 characters. (PCI DSS v1.1 8.5.10)

Passwords will contain both numeric and alphabetic characters. (PCI DSS v1.1 8.5.11)

The last 4 passwords will not be allowed for Location IDs during a password reset. (PCI DSS v1.1 8.5.12)

The Location ID password will be locked out after 5 bad attempts. (PCI DSS v1.1 8.5.13)

The lockout duration will be set to thirty minutes or until administrator enables the Location ID. (PCI DSS v1.1 8.5.14).

Appendix I – Contacts and Assistance Information

If a user encounters problems or needs assistance, contact the individual at your site responsible for ELIT. Whenever a user has trouble transmitting data to another location, check with your local ELIT support group first before assuming that something went wrong at the other end. Your local ELIT support group will work with the receiver site, if necessary, to resolve the problem.

For technical support while implementing the ELIT Client, please contact MFT Services via email, mft.services@dx.com or by telephone at 866 214 0108 option 1,1,2 (7:30 a.m.-4 p.m., Monday-Friday, Eastern time).

If you are unsure of who to contact please notify the BES Operations Service Desk for assistance at besoperations@dx.com or by telephone at 866-214-0108 option 1,1,1.

Document Version History

Version No.	Version Date	Revised By	Affected Section & Description of Change
1.0			Original.
1.1	09/01/2006		Introduction section – changed audit to auditable
1.2	10/09/2007		Enhanced with additional material
1.3	12/17/2008		Updated format to meet EDS/HP standards
1.4	06/03/2009		Added FROMDATA example to Sending Data, Site parameter section. Verbiage added to last sentence in 2.1
1.5	12/22/2009		Updated format to meet HP standards
1.6	09/24/2010		Latest enhancements for ACS-PCI usage. Added directory monitor section. Enhanced “Obtaining and installing the ELIT Client software” section. Added PORT prefix ‘E’ information.
1.7	09/30/2010		Updated for ipaccept parameter.
1.8	10/01/2010		Updated for logScriptStdout and logScriptStderr parameters.
1.9	05/25/2012		Modified the number of allowable monitored directories to 25.
2.0	08/27/2012		Added FileNameScrub property
2.1	09/16/2013		Removed insecure transfers of files.
2.2	11/18/2013		Updated for ELIT Client software version 4.3.2
2.3	10/13/2015	Julie Noelke	Change HP MFT to HPE MFT, @hp.com to @hpe.com, Hewlett-Packard to Hewlett Packard Enterprise, updated copyright.
2.4	11/06/2016	Julie Noelke	Updated to HPE template standard.
2.5	05/26/2017	Julie Noelke	Removed/updated references for HPE and Hewlett Packard Enterprise to DXC and DXC Technology.
2.6	06/06/2017	Julie Biafore	Updated copyright to reflect product creation date.
2.7	12/08/2017	Julie Biafore	Removed 248-365-2248 from Chapter 14.
2.8	02/13/2019	P. Szafranski	Updated for ELIT Client version 4.4.0.
2.9	12/21/2021	P. Szafranski/J. Biafore	Updated for ELIT Client version 4.5.0. Updated to DXC template format.